

GlowingStar “Yandaojie” AI Knowledge-Defense Platform

An evidence dossier on a Socratic, dialogue-based formative-assessment tool for primary-school learners

ACTION REQUIRED BEFORE SUBMISSION

Fields shown in **[gold]** are placeholders. Replace each with your own details, and confirm that every hyperlink is **publicly accessible and will remain live until the day of the Award Ceremony**, as required by the competition rules.

APPLICANT / SUBMITTING TEAM [Applicant name]	SCHOOL / ORGANISATION [School / Organisation]
AWARD & CATEGORY [Award name & category]	COUNTRY / REGION [Country / Region]
INITIATIVE GlowingStar · Yandaojie (研道阶)	EVIDENCE PERIOD Pilot to 1 June 2026
CONTACT [Email / phone]	DOCUMENT DATE 1 June 2026

This document accompanies the main application narrative. It catalogues the supporting materials offered as evidence and embeds the core data, frameworks and documentation inline so reviewers can assess the evidence base without leaving the dossier.

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1. Purpose & how to read this dossier

This pack strengthens the evidence base of our submission. **GlowingStar Yandaojie** (“研道阶”, literally a *staircase for inquiring into knowledge*) is an AI-powered *knowledge-defense* tool for primary-school pupils. Rather than grading multiple-choice answers, it runs a short, oral-style *defense*—much like a thesis viva—in which a teacher-modelled AI examiner reads a child’s reflection, then asks one probing open question at a time, adapting each follow-up to the pupil’s previous answer and diagnosing exactly which learning objectives are mastered and which are not.

The materials below map onto the categories the competition invites—*evaluation reports*, *implementation frameworks*, *policy/technical documents*, *testimonials*, *research references* and *classroom documentation*—and are designed to demonstrate **impact, sustainability and institutional embedding**. Quantitative claims are drawn directly from an anonymised export of live pilot data (see §6 and §10).

2. Inventory of supporting materials

#	Material	Type & what it evidences	Where to access
1	Pedagogical & implementation framework	Implementation framework — the reflect → defend → diagnose model and how it embeds in a 6th-grade lesson cycle (§3)	Embedded §3
2	Pilot impact dataset & analysis	Evaluation report — 50 learners, 140 defense sessions, 441 answered turns, objective-level mastery diagnoses (§4)	Embedded §4; raw data [dataset URL]

#	Material	Type & what it evidences	Where to access
3	Annotated classroom transcripts	Classroom documentation — real AI–pupil defense dialogues across Maths, Science and English (§5)	Embedded §5
4	Technical & methodology note	Technical document — model, prompt design, diagnostic schema and reproducible export pipeline (§6)	Embedded §6; repository [repo URL]
5	Sustainability & governance plan	Policy document — institutional embedding, deployment and data-governance posture (§7)	Embedded §7
6	Testimonials	Testimonials — teacher, pupil and parent voices (template to complete) (§8)	Embedded §8; video [link]
7	Research references	Research references — learning-science foundations for the design (§9)	Embedded §9
8	Live product / demo	Demonstration of the working tool	[product / demo URL]

3. Implementation framework — the pedagogical model

Each defense follows a fixed, teacher-designed structure so that the experience is consistent, fair and tightly aligned to the curriculum’s stated learning objectives.

3.1 The learner journey

1. **Reflect.** The pupil first writes a brief reflection against each lesson objective: *what I learned* and *what I’m still unsure about*. This primes metacognition and gives the examiner a starting point.
2. **Defend (5 rounds).** An AI examiner—modelled on “a rigorous but friendly Grade-6 teacher”—asks **exactly one open question per round**, for up to five rounds. Every question must explicitly build on something the child just said, target a named objective, and probe for weak reasoning, missing evidence or over-generalisation rather than reward recall.
3. **Diagnose.** After every round the system produces a structured diagnosis: which knowledge points the pupil has *mastered* and which are *not yet mastered*, with a one-line rationale each. This is the formative payload teachers act on.

3.2 Design guarantees (enforced in the examiner)

- Open questions only (“why / how / explain”)—no yes/no or multiple-choice prompts.
- One question, one question-mark per turn; no “and / also” compound questions.
- Questions are short enough for a Grade-6 pupil to answer in ~30 seconds.
- The examiner never answers for the pupil and never repeats an earlier question.
- Every question is tied to a specific curriculum objective; off-topic drift is disallowed.
- Subject-aware behaviour (e.g. for English, the examiner states explicitly when an answer must be given in English).

WHY “DEFENSE” INSTEAD OF A QUIZ

A quiz measures whether a final answer is right. A defense surfaces *how* a child reasons—and where the reasoning breaks. Because each follow-up is generated from the pupil's own words, the tool adapts to 50 different children without 50 hand-written question banks, and it hands the teacher an objective-level mastery map for the whole class.

4. Evidence of impact — pilot data

Source: anonymised export of real classroom pupils (numeric pupil IDs), generated 1 June 2026. Development/QA sessions are excluded. Figures describe a pilot and are reported transparently, including where engagement was low.

4.1 Headline figures

Pupils reached (pilot cohort)	50	Defense sessions run	140
Full 5-round defenses completed	85 (61%)	Answered defense turns	441
Objective-level diagnoses generated	353	Diagnosed as “mastered”	156 (44%)
Subjects covered	3	“I don’t know” / blank answers surfaced	96 (22%)

The 96 explicit “don’t-know” turns and the 197 “not-yet-mastered” diagnoses are treated as a feature, not a failure: the tool’s value is precisely in making hidden gaps visible to teachers early.

4.2 Engagement by subject

Mathematics		60 sess.
Science		45 sess.
English		35 sess.

Subject	Sessions	Completed (5 rounds)	Completion rate
Science — 《厨房里的物质变化》 (Matter changes in the kitchen)	45	40	89%
English — Unit 6 Summer Vacation	35	35	100%
Mathematics — 《可爱的小猫》 (coordinate scaling)	60	10	17%

The maths topic (scaling figures with coordinate pairs) drove the most attempts but the lowest completion—an honest, actionable signal that this concept needs more scaffolding before the defense, now a focus of our next iteration.

4.3 How far pupils get (rounds reached, all 140 sessions)

Reached round 5		85
Reached round 4		11
Reached round 3		13
Reached round 2		5
Reached round 1		3
Round 0 only		23

61% of sessions ran the full five-round defense—strong sustained engagement for an oral reasoning task at primary level. The 23 single-round sessions reflect first-touch exploration and abandoned attempts, useful for onboarding design.

5. Classroom documentation — annotated transcripts

Verbatim, anonymised dialogues from the pilot. They show the examiner building each question on the pupil's previous words and steering toward the objective. Original Chinese is preserved with English glosses.

SCIENCE · OBJECTIVE: CLASSIFY PHYSICAL VS. CHEMICAL CHANGE BY “IS A NEW SUBSTANCE FORMED?”

Q1 (AI): 你提到厨房里物质很多都在变化，请举一个具体的例子，并说说变化后是否产生了新物质？

“You said many things in the kitchen change—give one concrete example and say whether a new substance forms.”

A (pupil): 水变成了好喝的汤 — “Water turned into tasty soup.”

Q2 (AI): 你说水变成了好喝的汤，那汤里的水还是原来的水吗？

Builds directly on the child's answer to test the core concept.

A: 不是原来的了 因为有别的物质

Q5 (AI): 水在加热时会变成水蒸气，水蒸气是水产生的新物质吗？

A: 不是 状态不一样 — “No—it's a different state.” (correct distinction)

MATHEMATICS · OBJECTIVE: MEANING OF A COORDINATE PAIR AND SCALING ON GRID PAPER

Q (AI): 如果你在方格纸上看到一个点，写出的数对是 (5, 4)，那么 “5” 这个数表示什么意思呢？

A: 我不知道 — gap surfaced; flagged “not mastered”.

Q (AI, later round): 既然数对 (3, 2) 中的3是列、2是行，如果把两个数都乘2变成 (6, 4)，这个点会向什么方向移动？

Escalates from naming to reasoning about scaling—exactly the lesson objective.

6. Technology, methodology & data provenance

6.1 Architecture

- **Frontend:** Next.js 14 + Tailwind CSS pupil interface.
- **Backend:** FastAPI service with typed schemas; a defense-orchestration service builds each prompt from objectives, reflections and prior turns.
- **Reasoning model:** a large language model is prompted to return strict JSON containing the next question, the objective(s) it targets (with rationale) and the mastery diagnosis—making every AI decision auditable.
- **Persistence:** sessions, questions, turns, events and errors are stored for analysis.

6.2 Reproducible evaluation pipeline

A documented export script extracts every collection to (a) full-fidelity JSONL archives and (b) flattened CSV tables for Excel / R / SPSS / pandas, plus a `manifest.json` with row counts. Real pupils are

separated from QA accounts by a deterministic rule, so the impact figures in §4 are reproducible from the raw archive.

Exported table	Rows	Contents
sessions	179	One row per defense session
reflections	177	Pupil pre-defense reflections per objective
defense_questions	666	Generated questions + targeted objectives + diagnoses
defense_turns	620	Question/answer pairs with timestamps
events	1,435	Interaction telemetry
errors	232	Captured failures for reliability monitoring

7. Sustainability & institutional embedding

- **Curriculum-anchored, not bolt-on.** Defenses are authored against the exact objectives of standard Grade-6 lessons (Maths, Science, English), so the tool slots into existing teaching rather than competing with it.
- **Teacher in the loop.** The output is a class-level mastery map teachers use to plan re-teaching—the human teacher remains the decision-maker.
- **Maintainable engineering.** A single monorepo with isolated frontend/backend workspaces, linting and type-checking supports independent, low-risk iteration.
- **Reliability instrumentation.** Errors and events are logged and exported, giving an honest operational picture and a basis for continuous improvement.
- **Cost scalability.** One adaptive examiner replaces bespoke per-pupil question banks, so coverage scales to more pupils and subjects without proportional teacher workload.
- **Roadmap:** add scaffolding for low-completion topics (see Maths in §4.2), expand subject coverage, and longitudinal tracking of mastery gains across terms.

8. Testimonials (template to complete)

TO COMPLETE

Insert 2–4 short, attributable quotes with consent. Suggested mix: classroom teacher, head/curriculum lead, pupil, parent. Keep each to 2–3 sentences and add role + school.

Source	Quote	Attribution
Classroom teacher	["The defense showed me which children only memorised the steps..."]	[Name, role, school]
Pupil	["It kept asking me why, so I had to really think..."]	[Grade 6 pupil]
Parent / leader	[quote]	[Name, role]

9. Research references

The design draws on established learning science:

1. Black, P. & Wiliam, D. (1998). *Assessment and Classroom Learning*. *Assessment in Education*, 5(1), 7–74. — foundational case for formative assessment and feedback.
2. Vygotsky, L. S. (1978). *Mind in Society*. — the Zone of Proximal Development; questioning that adapts to the learner.
3. Chi, M. T. H. et al. (1994). Eliciting self-explanations improves understanding. *Cognitive Science*, 18(3). — the “explain why” effect the defense exploits.
4. Roediger, H. L. & Karpicke, J. D. (2006). The power of testing/retrieval practice. *Perspectives on Psychological Science*, 1(3).
5. Graesser, A. C. et al. (2005). AutoTutor and dialogue-based intelligent tutoring. *IEEE Trans. Education*. — precedent for conversational, Socratic AI tutoring.
6. OECD (2005). *Formative Assessment: Improving Learning in Secondary Classrooms*. — policy framing for embedding formative practice.

Replace or extend with the citation style your award requires; add DOIs/links as needed.

10. Data availability, privacy & ethics

- **Anonymisation.** Pilot pupils are identified only by non-identifying numeric IDs; the impact analysis contains no personal data.
- **Provenance.** All figures derive from a single dated export; the raw JSONL archive can be shared with reviewers under appropriate safeguards on request.
- **Transparency.** Every AI question records the objective it targets and its rationale, so decisions are inspectable rather than black-box.
- **Consent.** Confirm parental/school consent is in place before publishing any transcript, image or testimonial. **[Confirm consent status]**

11. Hyperlink index

COMPLIANCE REMINDER

Every link below must be **publicly accessible** and remain live for the full duration of the competition, including until the day of the Award Ceremony. Test each in a private/incognito window before submitting.

Resource	URL	Public?
Live product / demo	[https://...]	☐
Source repository	[https://...]	☐
Pilot dataset / methodology	[https://...]	☐

Resource	URL	Public?
Demo / classroom video	[https://...]	☐
Additional documentation	[https://...]	☐

GlowingStar · Yandaojie (研道阶) supporting-materials dossier — prepared 1 June 2026. Quantitative figures reproduced from an anonymised pilot data export. Placeholders in gold must be completed by the applicant prior to submission.